United States Patent	12408960
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Gabelberger; Josef et al.

Bone plate, bone plate system, and method of using the same

Abstract

A bone plate comprises a plate body and at least one tab. The plate body defines an inner body surface configured to face an underlying bone, an outer body surface opposite the inner body surface, and an outer side surface that extends between the inner body surface and the outer body surface. The outer side surface defines an outer perimeter of the plate body. The at least one tab includes a head and an arm that extends from the plate body to the head. The tab defines a tab aperture that extends through the head and is configured to receive a bone anchor. The bone anchor can be coupled with a screw hole in the nail. The arm is configured to deflect with respect to the plate body so as to move the head between a pre-fixation position and a fixation position spaced from the pre-fixation position.

Inventors: **Gabelberger; Josef (West Chester, PA), Cowens; David (West Chester, PA), McManus; Joshua (West Chester, PA), Keyer; Thomas (West Chester, PA), Santis; Catherine (Warsaw, IN), Khatchadourian; Robert (Lafayette Hill, PA), Van Citters; Peter (Phoenixville, PA), Bogle; David W. (Coatesville, PA), Sibole; Alexandra (Seattle, WA)**

Applicant: **DePuy Synthes Products, Inc.** (Raynham, MA)

Family ID: **78402169**

Assignee: **DePuy Synthes Products, Inc. (Raynham, MA)**

Appl. No.: **18/435146**

Filed: **February 07, 2024**

Prior Publication Data

Document Identifier	Publication Date
US 20240173057 A1	May. 30, 2024

Related U.S. Application Data

continuation parent-doc US 18060169 20221130 US 12016603 child-doc US 18435146
continuation parent-doc US 17071669 20201015 US 12011199 child-doc US 18060169

Publication Classification

Int. Cl.: **A61B17/80** (20060101); **A61B17/72** (20060101); **A61B17/88** (20060101)

U.S. Cl.:

CPC **A61B17/8085** (20130101); **A61B17/808** (20130101); **A61B17/8863** (20130101); A61B17/7233 (20130101); A61B17/8014 (20130101); A61B17/8052 (20130101); A61B17/8061 (20130101)

Field of Classification Search

CPC: A61B (17/8061); A61B (17/8014); A61B (17/808); A61B (17/8052); A61B (17/746); A61B (17/8085); A61B (17/8863); A61B (17/7233)

USPC: 606/70; 606/281; 606/291; 606/64; 623/21.18

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4398322	12/1982	Ewen	N/A	N/A
5122146	12/1991	Chapman et al.	N/A	N/A
5207682	12/1992	Cripe	N/A	N/A
5620456	12/1996	Sauer et al.	N/A	N/A
5704941	12/1997	Jacober et al.	N/A	N/A
6126359	12/1999	Dittrich et al.	N/A	N/A
6258095	12/2000	Lombardo et al.	N/A	N/A
6783529	12/2003	Hover et al.	N/A	N/A
7131974	12/2005	Keyer et al.	N/A	N/A
7357804	12/2007	Binder et al.	N/A	N/A
7549994	12/2008	Zander et al.	N/A	N/A
7618420	12/2008	Collazo	N/A	N/A
7621921	12/2008	Parker	N/A	N/A
7905883	12/2010	Bruecker et al.	N/A	N/A
8080010	12/2010	Schulz et al.	N/A	N/A
8080015	12/2010	Buettler et al.	N/A	N/A
8337533	12/2011	Raines et al.	N/A	N/A
8361077	12/2012	Keller	N/A	N/A
8486072	12/2012	Haininger	N/A	N/A
8668694	12/2013	Teeny	N/A	N/A
8753343	12/2013	Staeubli	N/A	N/A
8764752	12/2013	Buettler et al.	N/A	N/A
9066764	12/2014	Perez	N/A	N/A
9204912	12/2014	Price et al.	N/A	N/A
9308004	12/2015	Giersch et al.	N/A	N/A

9393064	12/2015	Roethlisberger et al.	N/A	N/A
D766434	12/2015	Dacosta	N/A	N/A
9439780	12/2015	Witt et al.	N/A	N/A
9463053	12/2015	Garino	N/A	N/A
9463054	12/2015	Mueckter	N/A	N/A
D780923	12/2016	Dacosta	N/A	N/A
9968389	12/2017	Garino	N/A	N/A
9993267	12/2017	Orsak et al.	N/A	N/A
10206724	12/2018	Williams	N/A	N/A
10258402	12/2018	Silva et al.	N/A	N/A
10327824	12/2018	Ricker et al.	N/A	N/A
10765462	12/2019	Penman et al.	N/A	N/A
D945623	12/2021	Daye	N/A	N/A
11413078	12/2021	Penman et al.	N/A	N/A
D977645	12/2022	Mason et al.	N/A	N/A
D977646	12/2022	Mason et al.	N/A	N/A
12016603	12/2023	Gabelberger	N/A	A61B 17/808
2003/0135212	12/2002	Y Chow	N/A	N/A
2006/0217722	12/2005	Dutoit	606/291	A61B 17/8085
2006/0235400	12/2005	Schneider	606/291	A61L 31/06
2009/0177240	12/2008	Perez	N/A	N/A
2009/0204121	12/2008	Cavallazzi et al.	N/A	N/A
2009/0306724	12/2008	Leither et al.	N/A	N/A
2010/0106196	12/2009	Erickson et al.	N/A	N/A
2010/0217327	12/2009	Vancelette et al.	N/A	N/A
2010/0256685	12/2009	Plecko	606/281	A61B 17/8061
2011/0190769	12/2010	Haininger	606/64	A61B 17/8061
2011/0264149	12/2010	Pappalardo et al.	N/A	N/A
2014/0094821	12/2013	Wagner et al.	N/A	N/A
2014/0107798	12/2013	Jeng et al.	N/A	N/A
2014/0378973	12/2013	Mueckter	606/64	A61B 17/746
2015/0366593	12/2014	Mebarak et al.	N/A	N/A
2016/0175112	12/2015	Pruvost et al.	N/A	N/A
2017/0056081	12/2016	Langdale et al.	N/A	N/A
2019/0133651	12/2018	Williams	N/A	N/A
2019/0262046	12/2018	Ricker et al.	N/A	N/A
2020/0214751	12/2019	Sixto et al.	N/A	N/A
2021/0212738	12/2020	Daye	N/A	N/A
2022/0031340	12/2021	Dogu et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	**Application Date**	**Country**	**CPC**
203619652	12/2013	CN	N/A

203935262	12/2013	CN	N/A
1398000	12/2003	EP	N/A
1398001	12/2003	EP	N/A
2030596	12/2008	EP	N/A
2349039	12/2015	EP	N/A
3122290	12/2018	EP	N/A
3084/MUM/2010	12/2012	IN	N/A
2011/017066	12/2010	WO	N/A
2015/144772	12/2014	WO	N/A
2017/198748	12/2016	WO	N/A
2018/072181	12/2017	WO	N/A
2020/168092	12/2019	WO	N/A

OTHER PUBLICATIONS

DePuy Synthes, Titanium Cannulated Retrograde/Antegrade Femoral Nail, Surgical Technique, Jun. 2017, 81 pages. cited by applicant

Interprosthetic and Peri-Implant Fractures: Principles of Operative Fixation and Future Directions, By Liporace et al., J Orthop Trauma, vol. 31, No. 5, May 2017, pp. 287-292. (www.jorthotrauma.com). cited by applicant

Intramedullary Nail and Plate Combination Fixation for Complex Distal Tibia Fractures: When and How? By Yoon et al., J Orthop Trauma, vol. 30, No. 11 Supplement, Nov. 2016 pp. S17-S21 (www.jorthotrauma.com). cited by applicant

Nail Plate Combination (NPC) Technique for Native and Periprosthetic Distal Femur Fractures A Technical Trick, Journal of Orthopaedic Trauma Publish Ahead of Print—DOI: 10.1097/BOT.0000000000001332, Liporace et al., pp. 1-26. cited by applicant

---

Primary Examiner: Shirsat; Marcela I.

Attorney, Agent or Firm: BakerHostetler

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This is a continuation of U.S. patent application Ser. No. 18/060,169 filed Nov. 30, 2022, which is a continuation of U.S. patent application Ser. No. 17/071,669 filed Oct. 15, 2020, the disclosure of each of which is hereby incorporated by reference as if set forth in its entirety herein.

TECHNICAL FIELD
(1) The present disclosure relates to systems, kits, assemblies, and methods for the placement and fixation of a bone plate against a bone for attachment to an intramedullary nail in a medullary canal of the bone.
BACKGROUND
(2) Intramedullary nails have long been used to treat fractures in long bones of the body such as fractures in femurs, tibias, and humeri. To treat such fractures, the intramedullary nail is inserted into a medullary canal of the long bone such that the nail extends across one or more fractures in the long bone to segments of the long bone that are separated by the one or more fractures. Bone anchors are then inserted through the bone and into the intramedullary nail on opposing sides of the fracture, thereby fixing the intramedullary nail to the bone. The intramedullary nail can remain in the medullary canal at least until the fracture is fused.

(3) The foregoing background discussion is intended solely to aid the reader. It is not intended to limit the innovations described herein. Thus, the foregoing discussion should not be taken to indicate that any particular element of a prior system is unsuitable for use with the innovations described herein, nor is it intended to indicate that any element is essential in implementing the innovations described herein.

SUMMARY

(4) The foregoing needs are met, to a great extent, by the system and method disclosed in the present application.

(5) Retrograde femoral nailing is often used for fixation of far distal fractures of the femur. When this occurs, often in combination with poor bone quality and/or in periprosthetic settings, additional bone fixation may be required. The proposed system links a small plate or washer to the distal end of the nail by means of a plurality of bone screws. In addition, there will be a plurality of peripheral bone screws that engage more bone than the nail's bone screws alone would, for improved fixation. Proper placement of the washer may present difficulty without the aid of an appropriate holding instrument. Attachment of a holding instrument may present difficulty to operating personnel, and/or the connection may not be rigid enough.

(6) The disclosure relates to a bone plate comprising a plate body and at least one tab. The plate body defines an inner body surface configured to face an underlying bone, an outer body surface opposite the inner body surface, and an outer side surface that extends between the inner body surface and the outer body surface. The outer side surface defines an outer perimeter of the plate body. The at least one tab includes a head and an arm that extends from the plate body to the head. The tab defines a tab aperture that extends through the head and is configured to receive a bone anchor body. The arm is configured to deflect with respect to the plate body so as to move the head between a pre-fixation position and a fixation position spaced from the pre-fixation position.

(7) According to another aspect of the present disclosure, a surgical kit for securing the bone plate to a bone is provided. The surgical kit comprises the bone plate and a bone plate adjustment tool. The bone plate adjustment tool is configured to be received within the tab aperture of the at least one tab member to transition the at least one tab member from the pre-fixation position to the fixation position.

(8) According to another aspect of the present disclosure, a bone plate holder for holding a bone plate during a surgical procedure is disclosed. The bone plate holder comprises a body and a shaft. The body defines a channel that extends through the body from a proximal surface of the body to a distal surface of the body. The shaft has a distal end that includes a coupling element configured to couple the shaft to the bone plate, and the shaft further having a proximal end opposite the distal end, wherein the proximal end includes a control element configured to control the coupling element to couple to the bone plate. The shaft is configured to extend through the channel, such that the coupling element extends out from the distal surface of the body.

(9) This summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description section. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. Furthermore, the claimed subject matter is not constrained to limitations that solve any or all disadvantages noted in any part of this disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The foregoing summary, as well as the following detailed description of illustrative embodiments of the implant of the present application, will be better understood when read in conjunction with the appended drawings. For the purposes of illustrating the implants of the present

application, there is shown in the drawings illustrative embodiments. It should be understood, however, that the application is not limited to the precise arrangements and instrumentalities shown. In the drawings:

(2) FIG. **1** illustrates a perspective view of a system according to one aspect having a bone plate holder supported and an aiming assembly that is attached to an intramedullary nail received in a medullary canal of a bone;

(3) FIG. **2** illustrates a top perspective view of a bone plate, according to an aspect of this disclosure;

(4) FIG. **3** illustrates a view of a side of the bone plate shown in FIG. **2**;

(5) FIG. **4** illustrates a top view of the bone plate shown in FIG. **2**;

(6) FIG. **5** illustrates a bottom view of the bone plate shown in FIG. **2**;

(7) FIG. **6**A illustrates a side cross-section view of the bone plate taken along line **6**A-**6**A in FIG. **4**;

(8) FIG. **6**B illustrates a side cross-section view of the bone plate taken along line **6**B-**6**B in FIG. **4**;

(9) FIG. **7** illustrates a close-up top view of at least one tab member of the bone plate shown in FIG. **2**;

(10) FIG. **8** illustrates a perspective view of a bone plate adjustment tool, according to an aspect of this disclosure;

(11) FIG. **9** illustrates a close-up view of a distal end of the bone plate adjustment tool shown in FIG. **8**;

(12) FIG. **10** illustrates a top perspective view of another aspect of a bone plate, according to an aspect of this disclosure;

(13) FIG. **11** illustrates a view of a side of the bone plate shown in FIG. **10**;

(14) FIG. **12** illustrates a top view of the bone plate shown in FIG. **10**;

(15) FIG. **13** illustrates a bottom view of the bone plate shown in FIG. **10**;

(16) FIG. **14** illustrates a perspective view of a bone plate holder, according to an aspect of this disclosure;

(17) FIG. **15** illustrates a perspective view of a body of the bone plate holder shown in FIG. **14**;

(18) FIG. **16** illustrates a cross-sectional view of the bone plate holder shown in FIG. **15** taken along line **16-16**;

(19) FIG. **17** illustrates a bottom view of the bone plate holder shown in FIG. **14**;

(20) FIG. **18** illustrates a perspective view of a shaft of the bone plate holder shown in FIG. **14**;

(21) FIG. **19** illustrates a top view of the shaft within the body of the bone plate holder shown in FIG. **14**;

(22) FIG. **20** illustrates a cross-sectional view of the shaft within the body shown in FIG. **19** taken along line **20-20**;

(23) FIG. **21** illustrates a perspective view of an intramedullary nail that is received in medullary canal of a bone, the intramedullary nail coupled to a handle shown in FIG. **1**;

(24) FIG. **22** illustrates a perspective view of a system comprising the intramedullary nail and handle as shown in FIG. **21** with an aiming guide and the bone plate holder shown in FIG. **14** fastened to the bone plate shown in FIG. **10**;

(25) FIG. **23** illustrates a side view of the system of FIG. **22** with the bone plate holder supporting the bone plate and a pair of guide sleeves;

(26) FIG. **24** illustrates a side view of the bone plate shown in FIG. **10** on a bone with the bone plate adjustment tool shown in FIG. **8** inserted into one of the tabs of the bone plate;

(27) FIG. **25** illustrates a perspective view of the bone plate shown in FIG. **2** fastened to the intramedullary nail; and

(28) FIG. **26** illustrates a perspective view of the bone plate shown in FIG. **10** fastened to the intramedullary nail.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

(29) The present disclosure can be understood more readily by reference to the following detailed

description taken in connection with the accompanying figures and examples, which form a part of this disclosure. It is to be understood that this disclosure is not limited to the specific devices, methods, applications, conditions or parameters described and/or shown herein, and that the terminology used herein is for the purpose of describing particular embodiments by way of example only and is not intended to be limiting of the scope of the present disclosure. Also, as used in the specification including the appended claims, the singular forms “a,” “an,” and “the” include the plural, and reference to a particular numerical value includes at least that particular value, unless the context clearly dictates otherwise.

(30) Certain terminology used in this description is for convenience only and is not limiting. The words “top”, “bottom”, “distal”, “proximal”, “inward”, “outward”, “inner”, “outer”, “above”, “below”, “axial”, “transverse”, “circumferential,” and “radial” designate directions in the drawings to which reference is made. The words “inner”, “internal”, and “interior” refer to directions towards the geometric center of the implant and/or implant adjustment tools, while the words “outer”, “external”, and “exterior” refer to directions away from the geometric center of the implant and/or implant adjustment tools. The words, “anterior”, “posterior”, “superior,” “inferior,” “medial,” “lateral,” and related words and/or phrases are used to designate various positions and orientations in the human body to which reference is made. The term “plurality”, as used herein, means more than one. When a range of values is expressed, another embodiment includes from the one particular value and/or to the other particular value. Similarly, when values are expressed as approximations, by use of the antecedent “about,” it will be understood that the particular value forms another embodiment. All ranges are inclusive and combinable. The terminology includes the above-listed words, derivatives thereof and words of similar import.

(31) As used herein, the term “substantially” and derivatives thereof, and words of similar import, when used to describe a size, shape, orientation, distance, spatial relationship, or other parameter includes the stated size, shape, orientation, distance, spatial relationship, or other parameter, and can also include a range up to 10% more and up to 10% less than the stated parameter, including 5% more and 5% less, including 3% more and 3% less, including 1% more and 1% less.

(32) An intramedullary nail is commonly secured to bone via at least one bone anchor, where each bone anchor extends directly into the surface of the bone and into a corresponding bone-anchor aperture in the intramedullary nail such that the bone anchor attaches to both the bone and the nail. However, as forces are exerted on the intramedullary nail, the attachment or bond between the bone anchor and the bone can weaken. This is especially true for patients whose bone at the bone anchor site is weakened due to osteoporosis or other bone conditions. To strengthen the attachment between the bone anchor and the bone, the bone anchor can be further secured to a bone plate that is positioned against the outer surface of the bone and secured to the bone via one or more additional bone anchors. For example, the bone plate can be positioned against the bone, and a first bone anchor can be inserted into an aperture in the plate, through the surface of the bone, and into the intramedullary nail, such that the first bone anchor attaches to the plate, the bone, and the intramedullary nail. Further, one or more other bone anchors can be inserted into the plate adjacent the first bone anchor such that the one or more other bone anchors terminate in the bone with or without passing into the intramedullary nail. The one or more other bone anchors provide additional fixation to the bone that can reduce loading on the first bone anchor.

(33) Referring to FIG. **1**, a system **10** is shown that is configured to position a bone plate **100** against a surface of a bone **70** as the bone plate **100** is fastened to the bone **70** and an intramedullary nail **60**. In general, the system **10** comprises a bone plate holder **400** (e.g. a bone-plate placement tool) that supports the bone plate **100** and angulates the bone plate **100** so as to align a surface of the bone plate **100** with a surface of the bone **70**. The system further comprises a bone plate adjustment tool **200** (see FIG. **8**) to facilitate fixation of the bone plate **100** to the bone **70**.

(34) The system **10** can further comprise one or more of the bone plates **100**, at least one bone

anchor **40** such as a bone screw, an aiming assembly **50**, and an intramedullary nail **60**. The intramedullary nail **60** is clongate generally along a superior-inferior direction SI and is sized to be received in a medullary canal of a long bone such as a femur, tibia, or humerus. The aiming assembly **50** releasably attaches to a proximal end of the intramedullary nail **60** and can comprise an aiming guide **80** and a handle **90**. The aiming assembly **50** can facilitate alignment of the bone plate **100** such that an axis As of the bone plate **100** is aligned with the bone plate holder **400** and the intramedullary nail **60**. For example, the axis of the bone plate **100** can be aligned with a bone-anchor opening that extends through the intramedullary nail **60**.

(35) Referring to FIGS. **2-7**, the bone plate **100** includes a plate body **101** having an inner body surface **102** (e.g. a bone-facing surface) and an outer body surface **104** opposite the bone-facing surface **102** along a transverse direction T. The body-facing surface **102** can have a curved shape, contoured shape, or other shape configured to align with the surface of the bone **70**. The bone plate **100** includes an outer side surface **103** that extends about an outer perimeter of the plate body **101**. The outer side surface **103** can have a first transverse side **106** and a second transverse side **108** opposite from one another. The first and second transverse sides **106** and **108** can extend from the bone-facing surface **102** to the outer body surface **104**. The outer side surface **103** can additionally or alternatively have a first lateral side **110** and a second lateral side **112** opposite from one another. The first and second lateral sides **110** and **112** can extend from the bone-facing surface **102** to the outer body surface **104**. The first and second lateral sides **110** and **112** can extend from the first transverse side **106** to the second transverse side **108**. It will be understood that embodiments of the disclosure are not limited to the specific bone plate shown in FIGS. **2-7**, and that alternative bone plates are contemplated (e.g. see bone plate **300** in FIGS. **10-13**).

(36) The bone plate **100** defines a first bone-anchor aperture **116** configured to be aligned with a shaft longitudinal axis A.sub.S1 (see FIG. **23**) when the bone plate **100** is fastened to the bone-plate placement tool **400**. Thus, the longitudinal axis A.sub.S1 can extend through the first bone-anchor aperture **116** when the bone plate **100** is fastened to the bone plate holder **400**. Further, the bone plate holder **400** is configured to align the longitudinal axis A.sub.S1 with the first bone-anchor aperture **116** over a full range of angles of the holder **400**, without impeding with a path of a bone anchor or drill bit with the bone plate holder **400**. The first bone-anchor aperture **116** can extend through the bone plate **100** from the outer body surface **104** to the bone-facing surface **102** so as to receive a bone anchor **40** (FIG. **1**) to attach the bone plate **100** to the bone **70**. The first bone-anchor aperture **116** can be threaded to receive a threaded head of a bone anchor. Further, the first bone-anchor aperture **116** can define variable-angle threading that permits a bone anchor to be inserted into the first bone-anchor aperture **116** at varying angles. Alternatively, the bone-anchor aperture **116** can be unthreaded.

(37) The bone plate **100** defines at least one additional bone-anchor aperture **118**, such as a plurality of additional bone-anchor apertures **118**. The at least one additional bone-anchor aperture **118** is spaced from the first bone-anchor aperture **116** such that the at least one additional bone-anchor aperture **118** is offset from (i.e., not aligned with) the shaft longitudinal axis A.sub.S1 when the bone plate **100** is fastened to the bone-plate placement tool **20**. The at least one additional bone-anchor aperture **118** extends through the bone plate **100** from the outer body surface **104** to the bone-facing surface **102**. At least one of the bone-anchor apertures **118** can be threaded to receive a threaded head of a bone anchor. Further, each bone-anchor aperture **118** can define variable-angle threading that permits a bone anchor to be inserted into the bone-anchor aperture **118** at varying angles. Alternatively, cach additional bone-anchor aperture **118** can be unthreaded.

(38) The bone plate **100** defines at least one, such as a plurality of, bone-plate fasteners **114** configured to engage a fastener of the bone plate holder **400** so as to fasten the bone plate **100** to the holder **400**. Each of the at least one bone-plate fasteners **114** can define an aperture configured to receive a corresponding fastener of the holder **400** so as to fasten the bone plate **100** to the holder **400**. For example, each fastener **114** can be configured to receive a projection that extends from the

bone-plate placement tool **20**. It will be understood that any other suitable fasteners can be used to releasably fasten the bone plate **100** to the bone plate holder **400**. For example, the fastener **114** can comprise a fixation element **350**, as further described below.

(39) The bone plate **100** further includes at least one tab member **130**. The at least one tab member **130** extends from the outer side surface **103** of the bone plate **100**. For example, the at least one tab **130** can extend from any one the first and second transverse sides **106** and **108** and the first and second lateral sides **110** and **112**, and/or can extend from multiple sides **106**, **108**, **110**, and **112** of the bone plate **100**. For example, FIGS. **2**, **4**, and **5** illustrate two tabs **130** that extend from the first lateral side **110**, and a single tab **130** that extends from the second transverse side **108**. Each of the at least one tabs **130** can extend in an outward direction (e.g. radial direction) from the outer side surface **103** of the bone plate **100**. In an aspect, each of the at least one tabs **130** extend in a direction from the outer side surface in a direction that is offset from the shaft longitudinal axis A.sub.S1 when the bone plate **100** is fastened to the bone-plate placement tool **20**. Each of the at least one tabs **130** can extend substantially parallel to each of the other tabs **130**, can be angularly offset from each of the other tabs **130**, or combinations of offset and parallel (e.g. some tabs **130** are parallel to each other, while other tabs **130** are angularly offset from other tabs **130**).

(40) Each tab **130** comprises an arm **132** and a head **134**. The arm **132** extends between the plate body **101** of the bone plate **100** and the head **134**. The arm **132** and the head **134** can be integrally formed as a single unitary piece. In an aspect, cach tab **130** and the body **101** of the bone plate **100** can be integrally formed as a single unitary piece. Alternatively, at least one of the tabs **130** can be coupled to (e.g. attachable) to the body **101** of the bone plate **100**.

(41) Each arm **132** can extend in an outward direction from the outer side surface **103** of the plate body **101**. Each arm **132** can extend substantially perpendicular to the outer side surface **103**. Alternatively, each arm **132** can extend at an angle other than substantially perpendicular to the outer side surface **103**. For example, with reference to FIG. **4**, each arm **132** that extends from the first lateral side **110** extends in the outward direction at an angle of between 0 and 90 degrees. The arm **132*a*** that extends from the second transverse side **108** extends in the outward direction at an angle of approximately 90 degrees. Each arm **132** can also extend in an inward or outward direction from the plate body **101**. For example, with reference to FIG. **4**, the arm **132*a*** that extends from the second transverse side **108** extends at least partially in an inward direction. The position and angle of each of the tabs **130** extending from the plate body **101** can depend on, for example, the anatomy of the patient, the size and/or shape of the intramedullary nail **60**, the location of corresponding apertures in the intramedullary nail **60**, or other factors.

(42) The head **134** of each tab **130** can define a tab aperture **136** that extends therethrough. With reference to FIGS. **6**A and **7**, the tab aperture **136*a*** extends through the head **134** from an upper opening **138** to a lower opening **140**. Each of the upper and lower openings **138** and **140** can be substantially perpendicular to a central aperture axis C.sub.A. The central aperture axis C.sub.A extends through a center of the tab aperture **136*a***. Alternatively, the upper and lower openings **138** and **140** can be angularly offset from a direction that is substantially perpendicular to the central aperture axis C.sub.A. For example, the lower opening **140** can be offset from the direction substantially perpendicular to the central aperture axis C.sub.A to align with an anatomy of the patient.

(43) The bone-anchor aperture **136*a*** further includes an inner surface **142** that extends about the central aperture axis C.sub.A. The inner surface **142** can define at least one column **144**. The at least one column **144** can extend about the central aperture axis C.sub.A from the upper opening **138** to the lower opening **140**. Alternatively, the at least one column **144** may extend about a portion of the inner surface **142**. For example, the at least one column **144** can extend from the lower opening **140** to a location between the upper and lower openings **138** and **140**. The at least one column **144** can include threads. In an aspect, the threads of the at least one column **144** can define variable-angle threading that permits a bone anchor to be inserted into the bone-anchor

aperture **136***a* at varying angles. Alternatively, the at least one column **144** of the bone-anchor aperture **136***a* can be unthreaded.

(44) The inner surface **142** can further define a receiving element **146**. The receiving element **146** is configured to receive the bone plate adjustment tool **200**, as further described below. The inner surface **142** can define a first cross-sectional dimension D.sub.1 (e.g. radial dimension extending from central aperture axis C.sub.A), and the receiving element **146** can define a second cross-sectional dimension D.sub.2 (e.g. radial dimension extending from central aperture axis C.sub.A) that is greater than the first cross-sectional dimension D.sub.1. The receiving clement **146** can comprise one or more recesses that include curved, lobed, rectangular, and/or other shapes. For example, the receiving element **146** can include a single or multi-lobe configuration, whereby each lobe of the receiving element **146** is spaced radially outward from the first cross-sectional dimension DI of the inner surface **142** relative to the central aperture axis C.sub.A. The number and spacing of the lobes of the receiving element **146** can correspond to a number and spacing of a connection element **202** of the bone plate adjustment tool **200**.

(45) The receiving element **146** can extend through the bone-anchor aperture **136***a* from the upper opening **138** to the lower opening **140**. Alternatively, the receiving element **146** can extend partially through the bone-anchor aperture **136***a* from the upper opening **138** to a location on the inner surface **142** between the upper and lower openings **138** and **140**. The receiving element **146** is configured to receive the bone plate adjustment tool **200** through the upper opening **138** and into the bone-anchor aperture **136***a*.

(46) The receiving element **146** can be spaced circumferentially about the inner surface **142**. For example, if the receiving element **146** includes 4 lobes, each lobe can be spaced approximately 90 degrees apart from each of the other lobes (e.g. cloverleaf configuration). Alternatively, each lobe of the receiving element **146** can be spaced at varying degrees about the inner surface, so long as the lobes correspond to the connection clement **202** of the bone plate adjustment tool **200**. The receiving element **146** can also circumferentially intersect with the at least one column **144** defined by the inner surface **142**, forming peripheral areas of thread relief arranged about the inner surface **142**. For example, for a receiving element **146** including 4 lobes spaced 90 degrees apart, the at least one column **144** can be positioned within the circumferential length of the 90 degrees between each lobe. In this example, the at least one column **144** can include 4 sections spaced circumferentially about the inner surface **142** configured to receive and engage a bone-anchor positioned within the bone-anchor aperture **136***a*. If the at least one column **144** includes a threaded section, the threaded section can threadedly engage a corresponding threaded section of a bone-anchor.

(47) Each of the bone-anchor apertures **136** defined by each tab **130** can be configured substantially similarly as each of the other bone-anchor apertures **136**. For example, the bone-anchor apertures **136** defined by the tabs **130** extending from the first lateral side **110** of the plate body **101** can have the same dimensions, threaded sections, receiving elements, or other features as the bone-anchor aperture **136***a* defined by the tab **130** extending from the second transverse side **108** of the plate body **101**. Alternatively, each of the bone-anchor apertures **136** can be configured differently from one or more of the other bone-anchor apertures **136**. For example, a cross sectional dimension (e.g. diameter and/or circumference) of the bone-anchor aperture **136***a* defined by the tab **130** extending from the second transverse side **108** of the plate body **101** can be greater than a cross sectional dimension of one or more of the tab apertures **136** defined by the tabs **130** extending from the first lateral side **110** of the plate body **101** and/or any of the other sides **106** and **112** of the plate body **101**. The particular configuration of the bone-anchor aperture **136** can correspond to the particular bone anchor being received within the aperture **136**.

(48) The at least one additional bone-anchor aperture **118** and the first bone-anchor aperture **116** can be configured in a substantially similar manner as the tab apertures **136** defined by the tabs **130**. For example, at least one or more of the additional bone-anchor apertures **118** and the first

bone-anchor aperture **116** can include the receiving element **146** and the at least one column **144**. Each receiving element **146** of the bone-anchor apertures **116** and **118** can be configured to correspond to the connection element **202** of the bone plate adjustment tool **200**.

(49) Each of the tabs **130** can be configured to flex and/or bend between multiple positions. Each of the tabs **130** can transition between a pre-fixation position, in which the head **134***a* is in a first position, and a fixation position, in which the head **134***a* is in a second position spaced from the first position. The flexing of the tabs **130** is discussed further below with reference to FIG. **24**. The tabs **130** can comprise a grade of stainless steel or other material adapted to plastically deform. In an aspect, the tabs **130** can comprise a carburized stainless steel to avoid galvanic corrosion caused between the contact between the tab **130** and the bone-anchor **40** positioned within the bone-anchor aperture **136**. For example, the intramedullary nail **60** can comprise titanium or a titanium alloy due to its low elastic modulus that encourages bone healing. If the bone-anchor **40** is made from either a stainless steel or a titanium, there can be a risk of galvanic corrosion on one end of the anchor **40** or another, where disparate metals contact each other. In an alternative aspect, the bone-anchor **40** can comprise a carburized stainless steel to avoid galvanic corrosion. It will be appreciated that other materials can be used to form the tabs **130**, the plate body **101**, and the anchors **40**, including, for example, cobalt chrome, cobalt chrome alloys, or still other materials.

(50) The arms **132** of each of the tabs **130** can include a substantially flat configuration. For example, each arm **132** can have a width that is greater than a thickness of the arm **132**. The flat configuration can facilitate bending, and in particular, can facilitate bending in a desired direction. For example, the arm **132** can bend in directions substantially perpendicular to the width of the arm **132**. The arm **132** can be aligned with respect to the plate body **101** of the bone plate **100** such that the arm **132** can be bent to substantially align a contour of the bottom surface of the tab **130** with a contour of the surface of the bone **70** of the patient during a surgical procedure.

(51) Referring to FIG. **6**B, the outer body surface **104** is spaced from the inner body surface **102** a first distance L.sub.1 along the transverse direction T so as to define a plate body **101** thickness. The arm **132***a* defines an inner arm surface **133** configured to face the underlying bone **70**. The arm **132***a* further defines an outer arm surface **135** opposite the inner arm surface **133** along the transverse direction T. The outer arm surface **135** is spaced from the inner arm surface **133** by a second distance L.sub.2 in the transverse direction T so as to define an arm **132***a* thickness. The first distance L.sub.1 that defines the plate body **101** thickness can be greater than the second distance L.sub.2 that defines the arm **132***a* thickness.

(52) The inner arm surface **133** is spaced from the outer arm surface **135** in an inward direction I that is oriented along the transverse direction T. In an aspect, the outer arm surface **135** is offset from the outer body surface **104** in the inward direction I. In an alternative or additional aspect, the outer arm surface **135** is offset with respect to the outer head surface **139** in the inward direction I. The inner arm surface **133** can be continuous with the inner head surface **137**. For example, the inner arm surface **133** may not be offset from the inner head surface **137** in the inward direction I such that the inner arm surface **133** extends adjacent to the inner head surface **137** without interruption. Alternatively, the inner arm surface **133** can be offset from the inner head surface **137** in the inward direction I.

(53) Referring to FIG. **7**, the arms **132***a* can extend from the plate body **101** to the head **134** along a central tab axis C.sub.T. Each arm **132***a* can be configured to twist about the central axis C.sub.T so as to move the head **134** from the pre-fixation position to the fixation position. Each arm **132** can include a first taper portion **129** and a second taper portion **131**. The first taper section **129** is located adjacent to the body **101**. The outer arm surface **135** within the first taper section **129** can taper in an outward direction. The outward direction being opposite the inward direction I. The inner arm surface **133** within the first taper section **129** can taper in the inward direction I. For example, the thickness of the arm **132***a* within the first taper section **129** can be greater than a thickness of the arm **132** along a length of the arm between the first taper section **129** and the

second taper section **131**.

(54) The second taper section **131** is located adjacent to the head **134***a*. The outer arm surface **135** within the second taper section **131** can taper in an outward direction. The inner arm surface **133** within the second taper section **131** can taper in the inward direction I. For example, the thickness of the arm **132***a* within the second taper section **131** can be greater than a thickness of the arm **132***a* along a length of the arm between the first taper section **129** and the second taper section **131**.

(55) Alternatively, or additionally, a width of the arm **132***a* within the first taper section **129** and the second taper section **131** can increase along its length from the arm **132***a* toward the body **101** and along its length from the arm **132***a* toward the head **134***a*, respectively. The width of the arm **132***a* defining a dimension that is substantially perpendicular to the inward direction I and the length of the arm **132***a* between the first and second taper sections **129** and **131**. The width of the arm **132***a* within the first taper section **129** can be greater than a width of the arm **132***a* along its length between the first and second taper sections **129** and **131**, and a width of the arm **132***a* within the second taper section **131** can be greater than a width of the arm **132***a* along its length between the first and second taper sections **129** and **131**.

(56) The head **134***a* defines an inner head surface **137** configured to face the underlying bone, and an outer head surface **139** opposite the inner head surface **137**. The outer head surface **139** is spaced from the inner head surface **137** by a third distance L.sub.3 so as to define a head **134***a* thickness. In an aspect, the third distance L**3** defining the head **134***a* thickness is substantially equal to the first distance Li defining the plate body **101** thickness. In another aspect, the third distance L.sub.3 defining the head **134***a* thickness is substantially equal to the second distance L.sub.2 that defines the arm **132***a* thickness. In an aspect, the thicknesses of the arm **132***a*, the head **134***a*, and the plate body **101** can be different.

(57) Referring to FIGS. **8** and **9**, the bone plate adjustment tool **200** includes the connection element **202**, a shaft **204**, and a handle **206**. The connection element **202** is positioned at the distal end **201** of the adjustment tool **200**. The shaft **204** extends between the connection element **202** and the handle **206**. In an aspect, the connection element **202** extends distally from a distal most end of the shaft **204**. The handle **206** extends from the shaft **204** toward a proximal end **203** of the adjustment tool **200**. The handle **206**, the shaft **204**, and the connection element **202** can be rigidly connected together such that movement and/or rotation of the handle **206** causes movement and/or rotation of the connection element **202**. The bone plate adjustment tool **200** is configured to engage and transition the tabs **130** from the pre-fixation position to the fixation position.

(58) The connection element **202** can comprise one or more protrusions **210** that include curved, lobed, rectangular, and/or other shapes. For example, the connection element **202** can include a single or multi-lobe configuration, whereby cach lobe (e.g. protrusion **210**) of the connection element **202** protrudes in a radially outward direction. Each lobe of a multi-lobe configuration can be spaced circumferentially about a center of the connection element **202**. The number and spacing of the protrusions **210** of the connection element **202** can correspond to a number and spacing of recesses of the receiving element **146** defined by the bone plate **100**.

(59) The connection element **202** is configured to engage the receiving element **146** to at least temporarily connect the adjustment tool **200** to the bone plate **100**. The connection between the adjustment tool **200** and the bone plate **100** can enable the adjustment tool to move and/or manipulate the bone plate **100**. For example, the adjustment tool **200** can bend and/or flex the tabs **130**. Additionally, the configurations of the receiving element **146** and the connection element **202** can reduce the risk of damage to the inner surface **142** of the bone-anchor aperture **136**. For example, if the inner surface **142** includes the at least one column **144** having a threaded region, the configurations of the receiving element **146** and the connection element **202** can reduce the risk of thread damage.

(60) It will be appreciated that more than one bone plate adjustment tools **200** can connect with the bone plate **100**. For example, a bone plate **100** having more than one receiving element **146** that

have different configurations (e.g. size, shape, etc.), multiple bone plate adjustment tools **200** can be used. In another example, a first bone plate adjustment tool **200** can engage a first bone-anchor aperture **136***a*, and a second bone plate adjustment tool **200** can engage a second bone-anchor aperture **136***a*. The first bone plate adjustment tool **200** can used to, for example, transition the tab **130** while the second bone plate adjustment tool **200** provides, for example, a counter torque to maintain the position of the bone plate **100**. Each bone plate adjustment tool **200** can comprise a a connection element **202** having a configuration (e.g. size, shape, etc.) that corresponds to a configuration of the respective receiving element **146**.

(61) FIGS. **10-13** illustrate an alternate aspect of a bone plate **300**, according to aspects of this disclosure. Portions of the alternate aspect of the bone plate **300** disclosed in FIGS. **10-13** are similar to aspects of the bone plate **100** described above in FIGS. **2-7** and those portions function similarly to those described above. The bone plate **300** includes a body **301** having a bone-facing surface **302** and an outer surface **304** opposite the bone-facing surface **302**. The bone plate **300** includes an outer side surface **303** that extends about an outer perimeter of the body **301**.

(62) The bone plate **300** defines a first bone-anchor aperture **316** that extends through the bone plate **300** from the outer surface **304** to the bone-facing surface **302** so as to receive the bone anchor **40** to attach the bone plate **300** to the bone **70**. The bone plate **300** defines at least one additional bone-anchor aperture **318**, such as a plurality of additional bone-anchor apertures **318**. The at least one additional bone-anchor aperture **318** is spaced from the first bone-anchor aperture **316**. The bone plate **300** further includes at least one tab member **330**.

(63) The bone plate **300** further includes the fixation element **350**. The fixation element **350** extends at least partially through the outer surface **304** and includes a recessed portion **352**, a first aperture **354**, and a second aperture **356** (e.g. a plate fixation element). The fixation element **350** is configured to receive a bone plate holder **400** within to connect the bone plate **300** to the bone plate holder **400**. The fixation element **350** can include fewer or more apertures **354** and **356** to align with corresponding elements of the bone plate holder **400**.

(64) The recessed portion **352** extends into the body **301** from the outer surface **304** to an inner recessed surface **358**. The first and second apertures **354** and **356** both extend from the inner recessed surface **358** into the body **301** toward the bone-facing surface **302**. In an aspect, one or both of the first and second apertures **354** and **356** extend through the body **301** to the bone-facing surface **302**. The first aperture **354** can be threaded to receive a corresponding threaded portion of a plate coupling element **432** of the bone plate holder **400**. The threads defined by the first aperture **354** can extend along an entire length of the first aperture **354**. Alternatively, the threads can extend along a partial length of the first aperture **354**. For example, the threads can extend from the inner recessed surface **358** toward the bone-facing surface **302**.

(65) The second aperture **356** is spaced apart from the first aperture **354** within the recessed portion **352**. The second aperture **356** can be unthreaded and configured to receive a corresponding portion (e.g. a holder fixation element **422**) of the bone plate holder **400**. The second aperture **356** can have a cylindrical shape, a conical shape, or other shape that corresponds to the holder fixation element **422** of the bone plate holder **400**.

(66) In an alternative aspect, one or both of the apertures **354** and **356** can be positioned within the recessed portion **352**. For example, one or both of the apertures **354** and **356** can extend into the body **301** from the outer surface **304** toward the bone-facing surface **302**.

(67) The recessed portion **352** includes an inner edge **353** extending about a perimeter of the recessed portion **352**. The inner edge **353** has a first end **355** spaced from a second end **357** in a plate lateral direction L. The second aperture **356** is positioned between the first aperture **354** and the first end **355** in the plate lateral direction L. The first aperture **354** is positioned between the second end **357** and the second aperture **356** in the lateral direction L. The inner edge **353** tapers in width, as measured in a direction substantially perpendicular to the plate lateral direction L, from a maximum width W.sub.1 of the inner edge **353** located toward the second end **357** to a minimum

width W.sub.2 located toward the first end **355**. A length between the maximum width W.sub.1 and the minimum width W.sub.2 along the inner edge **353** can be substantially linear.
(68) It will be appreciated that the fixation element **350** can include additional elements, for example, additional apertures, retentions elements, recesses/protrusions, or other elements. The fixation element **350** can also include other fixation configurations to connect to the bone plate holder **400**. For example, the fixation element **350** can include a snap-fit, friction-fit, or other type of fixation configuration.
(69) FIGS. **14-20** illustrate the bone plate holder **400**, according to aspects of this disclosure. The bone plate holder **400** includes a body **402** (e.g. holder body), a shaft **404**, and a grip member **406**. The grip member **406** is configured to be gripped by a surgeon performing a surgical procedure to align the bone plate holder **400** with the bone plate **100**, and to control the position of the bone plate **100** while the bone plate **100** is being fastened to the intramedullary nail **60**. The grip member **406** can connect to a handle connection portion **403** of the body **402**. The body **402** and the shaft **404** are configured to control connecting and disconnecting the bone plate holder **400** with the bone plate **100**.
(70) With reference to FIGS. **15** and **16**, the body **402** has an inner surface **410** that extends through the body **402** from a proximal surface **411** at a proximal end **412** of the body **402** to a distal surface **413** at a distal end **414** of the body **402**. The proximal end **412** is spaced from the distal end **414** in a proximal direction P. The inner surface **410** defines a shaft channel **416** that extends along a central channel axis C.sub.S through the body **402** from the proximal end **412** to the distal end **414**. The inner surface **410** further defines a shaft retention portion **418**. The shaft retention portion **418** is located toward the proximal end **412** of the body **402**. The shaft retention portion **418** can include a locking feature **420** to at least temporarily secure the shaft **404** within the shaft channel **416**. In an aspect, the locking feature **420** can include a threaded portion configured to engage with a corresponding body coupling element **434** of the shaft **404**.
(71) The body **402** further includes the holder fixation element **422** (e.g. a projection) configured to interface with the second aperture **356** of the bone plate **300**. The holder fixation element **422** can extend distally from the distal surface **413**. The holder fixation element **422** can have a substantially cylindrical shape, conical shape, or other shape. The shape and/or configuration of the holder fixation element **422** can correspond to a shape and/or configuration of the second aperture **356** of the bone plate **300**. The interface between the holder fixation clement **422** and the second aperture **356** facilitates alignment of the bone plate holder **400** with the bone plate **300**. It will be appreciated that the body **402** can include multiple holder fixation elements **422**, and the bone plate **300** can include multiple second apertures **356** that correspond to the multiple holder fixation elements **422**.
(72) Referring to FIG. **17**, the distal surface **413** at the distal end **414** of the body **402** includes an outer edge **415** extending about a perimeter of the distal end **414**. The outer edge **415** has a first end **417** spaced from a second end **419** in a lateral holder direction L′. The fixation clement **422** is positioned between a distal opening **421** of the shaft channel **416** defined by the distal surface **413** and the first end **417** in the lateral holder direction L′. The distal opening **421** is positioned between the second end **419** and the fixation element **422** in the lateral holder direction L′. The outer edge **415** tapers in width, as measured in a direction substantially perpendicular to the lateral holder direction L′, from a maximum width W′.sub.1 of the outer edge **415** located toward the second end **419** to a minimum width W′.sub.2 located toward the first end **417**. A length between the maximum width W′.sub.1 and the minimum width W′.sub.2 along the outer edge **415** can be substantially linear.
(73) A configuration of the distal end **414** of the body **402** can be configured to correspond to the recessed portion **352** of the fixation element **350** of the bone plate **300** so as to align the bone plate holder **400** with the bone plate **300** during connection of the holder **400** with the plate **300**. For example, the distal end **414** of the body **402** can be received within the recessed portion **352** such

that the plate lateral direction L aligns with the lateral holder direction L′ to at least partially rotationally lock the holder **400** with the plate **300** while the shaft **404** of the holder **400** is being secured to the fixation element **350** of the plate **300**. The maximum width W′.sub.2 and the minimum width W′.sub.1 of the outer edge **415** of the body **402** can be slightly less than the maximum width W.sub.2 and the minimum width W.sub.1 of the recessed portion **352**, respectively, such that an outer surface **405** of the body **402** contacts an inner surface **351** of the recessed portion **352**. The distal end **414** of the body **402** can be inserted into the recessed portion **352** of the plate **300** such that the distal surface **413** is substantially flush against the inner recessed surface **358**.

(74) With reference to FIG. **18**, the shaft **404** includes a control element **430** (e.g. a shaft grip member), the plate coupling element **432**, and the body coupling element **434**. The control element **430** is located at a proximal end **436** of the shaft **404**. The plate coupling element **432** is located at a distal end **438** of the shaft **404**. The body coupling element **434** is located between the plate coupling element **432** and the control element **430** along the shaft **404**. The shaft **404** is configured to be inserted into the shaft channel **416** of the body **402** through the proximal end **412**. The control element **430** can be positioned externally to the shaft channel **416**, such that the control element **430** extends out of the proximal surface **411** of the body **402** in the proximal direction P and is accessible by a surgeon during a procedure. A distal surface of the control element **430** can abut against the proximal surface **411** of the body **402** when the shaft **404** is positioned within the shaft channel **416**.

(75) The plate coupling element **432** can extend through an opening defined by the distal end **414** of the body **402**, such that the plate coupling element **432** extends distally out from the distal surface **413** of the body **402**.

(76) The body coupling element **434** is configured to engage with the locking feature **420** of the body **402**. In an aspect, the body coupling element **434** can include a threaded portion configured to engage with a corresponding threaded portion of the locking feature **420**. For example, during insertion of the shaft **404** into the shaft channel **416** of the body **402**, the body coupling element **434** can engage the locking feature **420** until the body coupling element **434** is positioned distal to the locking feature **420**. The position of the body coupling element **434** distal to the locking feature **420** can provide at least a temporary lock between the shaft **404** and the body **402**, such that the shaft **404** is at least partially restricted from exiting the shaft channel **416** through the proximal end **412** of the body **402**. For example, when the locking feature **420** and the body coupling element **434** are engaged and as the shaft **404** is rotated in a first direction of rotation (e.g. a clockwise direction), the body coupling element **434** moves distal to the locking feature **420** within the shaft retention portion **418**. When the body coupling element **434** is in the position distal of the locking feature **420** of the body **402**, the shaft **404** is then rotatable in the first direction of rotation without translating relative to the body **402**.

(77) The plate coupling element **432** is configured to connect with the first aperture **354** of the fixation element **350** of the bone plate **300**. The plate coupling element **432** can include a threaded portion **440** that engages a corresponding threaded section of the first aperture **354**. The shaft **404** can be rotated by rotating the control element **430** relative to the body **402** to threadedly connect the holder **400** to the bone plate **300**. When the plate coupling element **432** is connected to the first aperture **354**, and the holder fixation element **422** is connected to the second aperture **356**, the bone plate holder **400** is rigidly received within the plate **300** such that linear and rotational movement between the holder **400** and the platc **300** is substantially prevented. The rigid connection between the holder **400** and the plate **300** allows the surgeon to control the movement and placement of the bone plate **300** relative to the bone **70** of the patient during a surgical procedure.

(78) After the plate **300** is secured to the bone **70**, the bone plate holder **400** can be removed from the bone plate **300**. The shaft **404** can be disconnected from the first aperture **354** by rotating the control element **430** in a direction opposite to the direction of rotation for securing the plate

coupling element **432** to the first aperture **354**. After the plate coupling element **432** is disconnected, the holder **400** can be removed by retracting the distal end **414** of the body **402** from the recessed portion **352** and by retracting the holder fixation element **422** from the second aperture **356**.

(79) The control element **430** can include an engagement element **442** that defines a hexagonal or any alternatively shaped structure that can be engaged by a screw driving instrument, a protrusion element engageable by a wrench, or a cross hole configured to receive a cylindrical bar to rotate the control element **430** the bone plate holder **400** is being connected and disconnected to the plate **300**.

(80) With reference to FIGS. **21-26**, an example embodiment of a method of implanting an intramedullary nail **60** will be described. As shown in FIG. **21**, the handle **90** is coupled to the intramedullary nail **60**, and the intramedullary nail **60** is driven into the medullary canal of the bone **70** by the handle **90**. In FIG. **22**, the aiming guide **80** of the aiming assembly **50** is fastened to the handle **90**. The bone plate **300** is fastened to the bone plate holder **400** such that the bone plate **300** is configured to angulate with the plate holder **400** to position the bone plate **300** on the bone **70**. Although reference is made below with respect to bone plate **300**, it will be appreciated that the description also applies to bone plate **100** and/or other bone plate configurations.

(81) The patient can be oriented so that anatomic lateral shows condylar overlap (e.g. perfect anatomic lateral), and the nail **60** can be rotated to a desired position (e.g. obtain perfect circles). The nail **60** can be locked to the bone **70** (e.g. distal fragment) with a bone screw. In an aspect, the initial screw to lock the nail **60** to the bone **70** is a medial oblique screw. The screw substantially prevents motion of the nail **60** relative to the bone **70**.

(82) With reference to FIG. **23**, an incision can be made and the bone plate **300** is introduced to the bone **70**. The incision can include, for example, an 8 cm incision to connect first and second lateral guide sleeves **92** and **94**. The first guide sleeve **92** can extend along the shaft longitudinal axis A.sub.S1, and the second guide sleeve **94** can extend along a second longitudinal axis A.sub.S2. The first guide sleeve **92** aligns with the first bone-anchor aperture **316** and a first aperture in the intramedullary nail **60**, and the second guide sleeve **94** aligns with one of the additional bone-anchor apertures **318** and a second aperture in the nail **60**. A first drill bit (not shown) can be inserted into one of the first and second guide sleeves **92** and **94** to drill a first hole in the bone **70**. A second drill bit (not shown) can be inserted into the other of the first and second guide sleeves **92** and **94** to drill a second hole in the bone **70**. One of the first and second drill bits is removed from the bone, and a first bone-anchor screw (e.g. 5.0 VA locking screw) is inserted into the hole. The first bone-anchor can be positioned such that the bone-anchor is approximately 1 cm proud. After the first bone-anchor has been inserted, a second bone-anchor screw (e.g. 5.0 VA locking screw) is inserted into the other hole formed. The second bone-anchor can be positioned such that the bone-anchor is approximately 1 cm proud.

(83) After the first and second bone-anchors have been inserted, a lateral oblique bone-anchor screw can be inserted. After the first and second bone-anchors and the lateral oblique screw have been inserted, a lateral pressure can be applied to the plate **300** so the center and anterior portions of the plate **300** fit to the bone **70**. This can push one or more of the tabs **330** off the bone **70**. After the plate **300** is fit to the bone **70**, the first and second bone-anchors are tightened. The bone plate holder **400** can be removed from the plate **300** by rotating the control element **430** to release the plate coupling element **432** of the shaft holder **400** from the first aperture **354** of the bone plate **300**. The anterior locking screws can be inserted through the additional bone-anchor apertures **318**. In an aspect, holes can be pre-drilled in the bone prior to inserting the anterior locking screws.

(84) With reference to FIG. **24**, the bone plate adjustment tool **200** can manipulate each of the at least one tab members **330** so that the posterior portions of the plate **300** fit to the bone **70**. For example, the connection element **202** of the adjustment tool **200** is inserted into the respective aperture **336** of the tab **330**, and a pressure (e.g. a deflection force) can be applied to the respective

head **334** of the tab **330** to cause the arm **332** to deflect the head **334** from the pre-fixation position to the fixation position. For example, the arm **332** can deflect with respect to the plate body **301** in response to a force so as to move the head **334** between the pre-fixation position and the fixation position. The force applied to the arm **332** to transition the head between the pre-fixation position and the fixation position can be insufficient to cause the plate body **101** to deflect. After the head **334** is transitioned to the fixation position, the inner head surface **137** abuts the underlying bone. In an aspect, the arm **332** is deformable such that the fixation position of the head **334** is offset in the inward direction I with respect to the pre-fixation position of the head **334**.

(85) During transition, the trajectory of the screw through the respective aperture **336** should be considered to ensure the screw is securely located within the bone **70**. After the tabs **330** have been transitioned to their respective fixation position, posterior bone- anchor screws can be inserted through each aperture **336** and into the bone **70**. After each of the bone-anchors has been inserted, a final torque can be applied to the bone-anchors.

(86) Although the disclosure has been described in detail, it should be understood that various changes, substitutions, and alterations can be made herein without departing from the spirit and scope of the invention as defined by the appended claims. Additionally, any of the embodiments disclosed herein can incorporate features disclosed with respect to any of the other embodiments disclosed herein. Moreover, the scope of the present disclosure is not intended to be limited to the particular embodiments described in the specification. As one of ordinary skill in the art will readily appreciate from that processes, machines, manufacture, composition of matter, means, methods, or steps, presently existing or later to be developed that perform substantially the same function or achieve substantially the same result as the corresponding embodiments described herein may be utilized according to the present disclosure.

Claims

1. A system comprising: a bone plate including: a plate body that defines a bone-facing body surface configured to face a bone and an outer body surface opposite the bone-facing body surface, wherein the bone plate defines a plurality of threaded bone anchor apertures that extend through the plate body from the outer body surface to the bone-facing body surface and are each configured to receive a respective bone anchor, the plate body is devoid of flexible arms, and the plate body defines an outermost plate body perimeter that extends about the plate body at a location between the bone-facing body surface and the outer body surface; first and second tabs each including a respective flexible arm that extends from the plate body along a respective length direction to a respective head that is supported by the respective flexible arm, and a respective tab aperture that extends through the respective head and is configured to receive a respective bone anchor, wherein each of the first and second tabs terminates at the respective head, wherein each of the flexible arms defines a respective inner bone-facing surface and a respective outer surface opposite the respective inner bone-facing surface along a thickness that extends along a thickness direction, wherein each of the flexible arms defines respective opposed sides that are opposite each other along a width that extends along a width direction, and the width is greater than the thickness along respective entireties of the flexible arms, and the sides are all straight and linear, wherein the opposed sides of the flexible arm of the first tab are spaced from each other along the outermost body perimeter along the width direction, and the opposed sides of the flexible arm of the second tab are spaced from each other along the outermost body perimeter along the width direction, wherein both of the respective opposed sides of the first flexible arm extend directionally away from both of the respective opposed sides of the second flexible arm from the plate body to the respective head; and an intramedullary nail configured to be inserted into medullary canal of the bone, wherein the intramedullary nail is configured to receive the bone anchors so as to fix the intramedullary nail to the plate body.

2. The system of claim 1, wherein each of the threaded bone anchor apertures defines respective pairs with each other of the threaded bone anchor apertures, and for each respective pair of threaded bone anchor apertures, a straight line that extends through respective centers of the bone anchor apertures of the pair divides the plate body into a first plate body side and a second plate body side, and the tabs extend from only one of the first and second plate body sides, such that no tabs extend out from the other of the first and second plate body sides.
3. The system of claim 1, wherein: the plate body defines a first direction, a second direction perpendicular to the first direction, and a third direction opposite the second direction and perpendicular to the first direction, the bone plate defines a bone plate aperture, wherein the bone plate aperture is disposed between first and second ones of the threaded bone anchor apertures with respect to the first direction, such that a straight line that passes through the bone plate aperture and is oriented along the first direction also passes through the first and second ones of the threaded bone anchor apertures, wherein the plurality of threaded bone anchor apertures further includes third and fourth ones of the plurality of threaded bone anchor apertures that are offset with respect to the bone plate aperture in the second direction, such that a second straight line that is oriented along the first direction and extends through the third one of the plurality of fixation holes also passes through the fourth one of the plurality of fixation holes; and each of the respective heads is offset from the plate body in the third direction.
4. The system of claim 1, wherein the threaded bone anchor apertures comprise variable angle threading.
5. The system of claim 1, wherein the bone plate further comprises a bone plate aperture that extends through the plate body that is configured to receive a corresponding fastener of a bone plate holder.
6. The system of claim 5, wherein the bone plate aperture is threaded.
7. The system of claim 6, further comprising a second bone plate aperture that is configured to receive a respective feature of the bone plate holder.
8. The system of claim 7, wherein the second bone plate aperture is unthreaded.
9. The system of claim 1, further comprising the bone plate holder.
10. The system of claim 1, wherein the plate body has a plate body thickness that extends from the bone-facing body surface to the outer body surface, and an entirety of the plate body thickness is greater than the thickness of each of the flexible arms.
11. A system comprising: a bone plate including: a plate body that defines a bone-facing body surface configured to face a bone and an outer body surface opposite the bone-facing body surface, wherein the bone plate defines a plurality of threaded bone anchor apertures that extend through the plate body from the outer body surface to the bone-facing body surface and are each configured to receive a respective bone anchor, and the plate body is devoid of flexible arms; and first and second tabs each including a respective flexible arm that extends from the plate body, a respective head supported by the respective flexible arm, and a respective tab aperture that extends through the respective head and is configured to receive a respective bone anchor, wherein each of the flexible arms defines a respective inner bone-facing surface and a respective outer surface opposite the respective inner bone-facing surface along a respective thickness that extends along a thickness direction, and wherein each of the flexible arms defines respective opposed sides that are opposite each other along a width that extends along a width direction, and the width is greater than the thickness along respective entireties of the flexible arms, and wherein both of the respective opposed sides of each of the first and second flexible arms extend directionally away, along respective entireties of their lengths from the plate body to the respective head, from the other of the first and second flexible arms; and an intramedullary nail configured to be inserted into medullary canal of the bone, wherein the intramedullary nail is configured to receive the bone anchors so as to fix the intramedullary nail to the plate body.
12. The system of claim 11, wherein each of the threaded bone anchor apertures defines respective

pairs with each other of the threaded bone anchor apertures, and for each respective pair of threaded bone anchor apertures, a straight line that extends through respective centers of the bone anchor apertures of the pair divides the plate body into a first plate body side and a second plate body side, and the tabs extend from only one of the first and second plate body sides, such that no tabs extend out from the other of the first and second plate body sides.

13. The system of claim 11, wherein: the plate body defines a first direction, a second direction perpendicular to the first direction, and a third direction opposite the second direction and perpendicular to the first direction, the bone plate defines a bone plate aperture, wherein the bone plate aperture is disposed between first and second ones of the threaded bone anchor apertures with respect to the first direction, such that a straight line that passes through the bone plate aperture and is oriented along the first direction also passes through the first and second ones of the threaded bone anchor apertures, wherein the plurality of threaded bone anchor apertures further includes third and fourth ones of the plurality of threaded bone anchor apertures that are offset with respect to the bone plate aperture in the second direction, such that a second line that is oriented along the first direction and extends through the third one of the plurality of fixation holes also passes through the fourth one of the plurality of fixation holes; and each of the respective heads is offset from the plate body in the third direction.

14. The system of claim 11, wherein the bone plate further comprises a bone plate aperture that extends through the plate body that is configured to receive a corresponding fastener of a bone plate holder.

15. The system of claim 14, wherein the bone plate aperture is threaded.

16. The system of claim 15, further comprising a second bone plate aperture that is configured to receive a respective feature of the bone plate holder.

17. The system of claim 16, wherein the second bone plate aperture is unthreaded.

18. The system of claim 11, further comprising the bone plate holder.

19. The system of claim 11, wherein the opposed sides of the first and second flexible arms extend straight and linearly from the plate body to the respective heads.

20. The system of claim 11, wherein the plate body defines an outermost plate body perimeter that extends about the plate body at a location between the bone-facing body surface and the outer body surface, and the width direction extends along the outermost plate body perimeter at an interface between the outermost plate body perimeter and the arms.